

projectile stage



$$\vec{r}_m = x \hat{n}_1 + y \hat{n}_2$$

$$\vec{v}_m = \dot{x} \hat{n}_1 + \dot{y} \hat{n}_2$$

$$T = \frac{1}{2} m |\vec{v}_m|^2 = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2)$$

$$V = mgy$$

$$L = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \dot{y}^2 + mgy$$

x:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) + \frac{\partial L}{\partial x} = 0$$

$\downarrow$ $\downarrow$
 mx 0

$$x_0 = 0$$

$$m\ddot{x} = 0 \quad \therefore \ddot{x} = 0$$

$$\dot{x} = v_{x0} \quad x(t) = v_{x0}t + x_0$$

$$\downarrow: \quad m\ddot{y} + mg = 0$$

$$\ddot{y} = -g$$

$$\text{if } \ddot{y} = -g$$

$$\frac{dy}{dt} \left(\frac{dy}{dt} \right) = -g$$

$$\dot{y}(t) = -gt$$

$$y(t) = -\frac{gt^2}{2} + y_0$$

Projectile stage

$$\ddot{x} = 0$$

$$\dot{x} = v_{x0}$$

$$x_{TD} = v_{x0} t_{TD}$$

$$\dot{x}_{TD} = v_{x0}$$

$$x(t) = v_{x0} t + x_0$$

$$\ddot{y} = -g$$

$$\dot{y}(t) = -gt$$

$$y(t) = y_0 - \frac{g}{2} t^2$$

assume
 θ_{TD} is
constant
in air

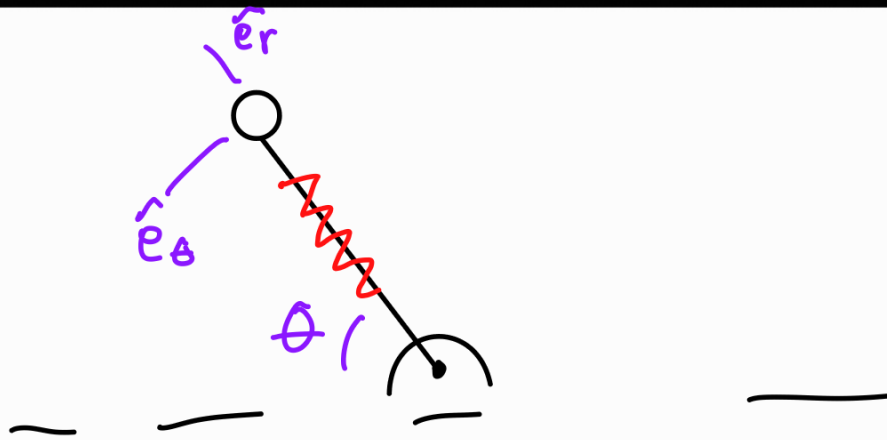
$$\vec{r}_{foot} = (x - r_0 \cos \theta_{TD}) \hat{n}_1 + (y - r_0 \sin \theta_{TD}) \hat{n}_2$$

$$y_{TD} = r_0 \sin \theta_{TD} \quad \dot{y}_{TD} = -\sqrt{2g(y_A - r_0 \sin \theta_{TD})}$$

$$t_{TD} = \sqrt{\frac{2(y_0 - r_0 \sin \theta_{TD})}{g}}$$

Spring stage

$r \dot{\theta}^2$
 $r \ddot{\theta}$



$$\vec{r}_m = r \hat{e}_r$$

$$\frac{d\vec{r}_m}{dt} = \frac{d\vec{r}}{dt} + \vec{\omega}_{E/N} \times \vec{r}_m$$

$$\dot{r} \hat{e}_r + \dot{\theta} \hat{e}_3 \times r \hat{e}_r$$

$$\vec{v}_m = \dot{r} \hat{e}_r + r \dot{\theta} \hat{e}_\theta$$

$$|\vec{v}_m|^2 = \dot{r}^2 + r^2 \dot{\theta}^2$$

$\therefore$

$$T = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\theta}^2$$

$$V = mgr \sin \theta + \frac{1}{2} k (r - r_0)^2$$

$$L = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\theta}^2$$

$$- mgr \sin \theta - \frac{1}{2} k (r - r_0)^2$$

$$r: \quad \frac{\partial L}{\partial r} : m r \dot{\theta}^2 - mg \sin \theta - k(r - r_0)$$

$$\frac{\partial L}{\partial \dot{r}} : m \dot{r} \xrightarrow{\partial/\partial t} m \ddot{r}$$

$\therefore$

$$m \ddot{r} - m r \dot{\theta}^2 + mg \sin \theta + k(r - r_0) = 0$$

$$\theta: \quad \frac{\partial L}{\partial \theta} : -mgr \cos \theta$$

$$\frac{\partial L}{\partial \dot{\theta}} : m r^2 \dot{\theta} \xrightarrow{\partial/\partial t} m r^2 \ddot{\theta} + 2 m r \dot{r} \dot{\theta}$$

$$m r^2 \ddot{\theta} + 2 m r \dot{r} \dot{\theta} + mgr \cos \theta = 0$$

All coords in L , None ignorable

$\vec{r}_n$ has no t dependancy

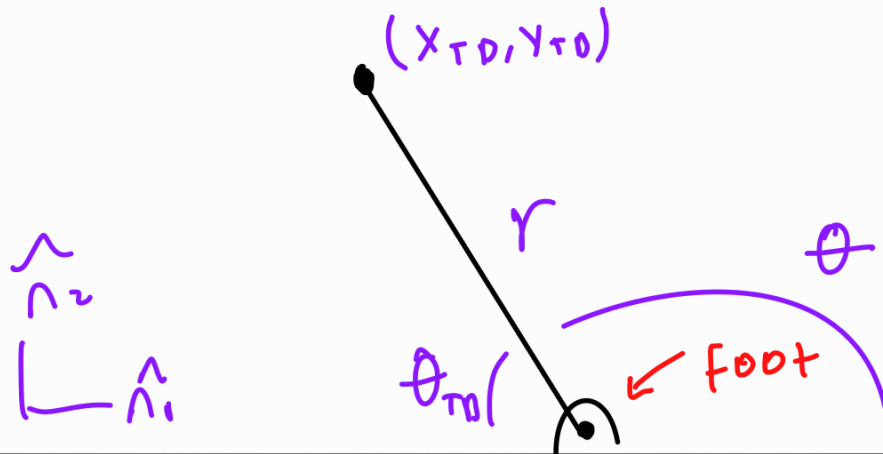
scleronomous,

T is quadratic

$$H = T + V$$

$\frac{\partial L}{\partial t} = 0$, E is conserved

Touch down



$$x_f = x_{TD} + r \cos \theta_{TD}$$

$$\vec{r}_m = x_{TD} \hat{n}_1 + y_{TD} \hat{n}_2$$

$$\vec{r}_f = (x_{TD} + r_0 \cos \theta_{TD}) \hat{n}_1 + 0 \hat{n}_2$$

$$\vec{p} = \vec{r}_m - \vec{r}_f$$

$$= -r \cos \theta_{TD} \hat{n}_1 + y_{TD} \hat{n}_2$$

$$r = |\vec{p}| = \sqrt{r_0^2 \cos^2 \theta_{TD} + r_0^2 \sin^2 \theta_{TD}}$$

$$r = r_0$$

$$\theta = \arctan \left| \frac{r_0 \sin^2 \theta}{r_0 \cos \theta} \right|$$

$$\vec{e}_r = \cos\theta \hat{n}_1 + \sin\theta \hat{n}_2$$

$$\vec{e}_\theta = -\sin\theta \hat{n}_1 + \cos\theta \hat{n}_2$$

$$\vec{v} = \dot{x}_{TD} \hat{n}_1 + \dot{y}_{TD} \hat{n}_2$$

$$\vec{v} = \dot{r} \hat{e}_r + r \dot{\theta} \vec{e}_\theta$$

$$\dot{r} = \vec{v} \cdot \vec{e}_r = \dot{x}_{TD} \cos\theta + \dot{y}_{TD} \sin\theta$$

$$r \dot{\theta} = \vec{v} \cdot \vec{e}_\theta = -\dot{x}_{TD} \sin\theta + \dot{y}_{TD} \cos\theta$$

$$\dot{\theta} = \frac{-\dot{x}_{TD} \sin\theta + \dot{y}_{TD} \cos\theta}{r}$$

$$m\ddot{r} - m r \dot{\theta}^2 + mg \sin \theta + k(r - r_0) = 0$$

$$m r^2 \ddot{\theta} + 2 m r \dot{r} \dot{\theta} + m g r \cos \theta = 0$$

$$\ddot{r} = r \dot{\theta}^2 - g \sin \theta - \frac{k}{m} (r - r_0)$$

$$\ddot{\theta} = -\frac{2}{r} \dot{r} \dot{\theta} - \frac{g}{r} \cos \theta$$